Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 03-Apr-2026

---

1. Administrative & Study Identification

Full Study Title: Two-period Crossover Study to Demonstrate the Comparability of Pharmacokinetics of Subcutaneous Ianalumab Between 2mL Auto-injector/2mL PFS with 1mL Pre-filled Syringe in Adult Participants With Autoimmune Disease **Short Title/Acronym:** Ianalumab PK Device Comparability Study **Protocol Number:** CVAY736A2202 **NCT Number:** NCT06293365 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Ianalumab (VAY736) **Phase of Development:** Phase 2 **Medical Monitor:** Novartis Clinical Operations Lead

---

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To demonstrate the comparability of ianalumab exposure (pharmacokinetics) following the subcutaneous (s.c.) administration of one injection of 300 mg/2 mL auto-injector (AI) or 2 mL pre-filled syringe (PFS) versus two injections of 150 mg/1 mL PFS. **Secondary Objectives:** To evaluate the safety and tolerability of ianalumab following the s.c. administration of both devices in participants with active autoimmune diseases.

2.2 Background & Rationale To provide a more convenient and patient-friendly administration method for patients requiring B-cell depletion therapy for active autoimmune diseases, including Rheumatoid Arthritis (RA), Sjögren's Disease (SjD), and Systemic Lupus Erythematosus (SLE). Bridging pharmacokinetic (PK) data is required to establish that a single 2 mL injection (via AI or PFS) yields comparable systemic exposure to the standard two 1 mL PFS injections.

---

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Crossover) **Blinding:** Open-label **Randomization:** 1:1, Two-period, two-sequence crossover

3.2 Planned Enrollment & Duration Target N: 155 evaluable participants **Number of Sites:** Global sites across Europe, Canada, US, and Rest of World (approx. 37 locations) **Estimated Study Start:** 02-Jul-2024 **Estimated Primary Completion:** 05-Jun-2025

---

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female patients aged 18 to 70 years (inclusive). 2. Diagnosed with RA, SjD, and/or SLE with active disease that may benefit from B-cell depletion therapy, as determined by the investigator. 3. Body weight ≥ 35 kg and ≤ 150 kg; BMI within the range of 18 - 35 kg/m^2. 4. Participants receiving protocol-allowed standard of care (SoC) must be on stable doses for 4 weeks prior to first dosing.

4.2 Exclusion Criteria 1. Pregnant or nursing (lactating) women, or women of child-bearing potential refusing to use highly effective contraception. 2. Active viral, bacterial, or other infections requiring systemic treatment at the time of screening/ baseline, or a history of recurrent clinically significant infections. 3. Plans for administration of live vaccines during the study period. 4. Uncontrolled co-existing serious disease or use of prohibited therapies.

---

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ianalumab 300 mg administered monthly (Sequence 1: 2 x 1 mL PFS then 1 x 2 mL AI/ PFS; Sequence 2: 1 x 2 mL AI/PFS then 2 x 1 mL PFS). **Route of Administration:** Subcutaneous (s.c.) **Duration of Treatment:** 24 weeks (Two 12-week treatment periods), with an optional extended treatment period up to Week 72.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of standard of care (SoC) medications. **Prohibited:** Prohibited immunosuppressive/biologic therapies and live vaccines during the study period.

---

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to -28 Days	Informed Consent, Eligibility Assessment, Baseline Labs
Treatment Period 1	Week 0 to 12	Randomization, First Sequence Dosing, PK Sampling
Treatment Period 2	Week 12 to 24	Treatment Switch (Crossover), PK Sampling
Extended Treatment	Week 24 to 72	Monthly Dosing with AI or PFS (Cohort dependent)
End of Study / Follow-Up	Week 72 to 176	End of Treatment Visit (Wk 72), Mandatory Follow-Up (Wk 88), Conditional Follow-Up until B-cell recovery or up to 2 years

---

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Area under the curve to the end of a dosing interval at steady-state ($AUCtau$) and Maximum observed serum drug concentration ($Cmax$) for ianalumab. **Measurement Method:** Pharmacokinetic (PK) blood sampling evaluated over a dosing interval after the 3rd dose (Weeks 8-12) and 6th dose (Weeks 20-24).

7.2 Secondary & Exploratory Endpoints Safety and tolerability assessments, including Adverse Events (AEs) and Serious Adverse Events (SAEs) tracking. B-cell recovery tracking (defined as $CD19+$ B-cell counts returning to ≥ 50 cells/μL or $\ge 80\%$ of baseline value).

---

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite clinical visits, monitoring injection site reactions, and systemic tolerability to the s.c. administration. **Serious Adverse Events (SAEs):** Reported within standard regulatory timelines (typically 24 hours). **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by Novartis Pharmaceuticals.

---

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Set for exposure comparability analysis; Safety Set for tolerability endpoints. **Sample Size Justification:** Target sample size of $N=155$ designed to provide adequate statistical power to demonstrate pharmacokinetic comparability (bioequivalence boundaries) across the different device cohorts. **Handling of Missing Data:** Standard PK modeling procedures and mixed-effects analysis for crossover designs.

---

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote source data verification according to Phase 2 standard clinical trial guidelines. **Audit Trail:** Secure eCRF systems with comprehensive audit trails for PK sampling and administration records.

---

11. Study Identifiers & Governance

Primary ID: CVAY736A2202 **Registry Number:** NCT06293365 **Sponsor/ Lead Organization:** Novartis Pharmaceuticals **Study Title:** Two-period Crossover Study to Demonstrate the Comparability of Pharmacokinetics of Subcutaneous Ianalumab Between 2mL Auto-injector/2mL PFS with 1mL Pre-filled Syringe in Adult Participants With Autoimmune Disease **Official Regulatory Title:** A

Randomized, Two-period Crossover Study to Demonstrate the Comparability of Pharmacokinetics of Subcutaneous Ianalumab Between 2mL Auto-injector/2mL PFS with 1mL Pre-filled Syringe in Adult Participants With Autoimmune Disease

---

12. Clinical Framework (Autoimmune Specific)

Primary Condition: Rheumatoid Arthritis, Sjögren's Disease, Systemic Lupus Erythematosus **Diagnosis Threshold:** Active disease presentation as determined by investigator **Medical Methodology:** B-cell depletion therapy **Optimization Target:** Demonstration of equivalent PK exposure via updated delivery devices (AI vs PFS)

---

13. Study Procedures & Methodology

Management Pathway: Pharmacokinetic device bridging via crossover design **Education Protocol (HCP):** Site administration guidelines and device handling protocol **Education Protocol (Patient):** Protocol for reporting injection site reactions and symptom monitoring **Quality Audit Mechanism:** Drug accountability and precise PK blood draw timing adherence

---

14. Observation & Follow-Up Schedule

Total Duration: Up to 176 weeks (Screening + Treatment + Conditional Follow-Up) **Onsite Visit Cadence:** Frequent visits across Weeks 0, 12, 24, and 72 (End of Treatment) **Remote Monitoring Frequency:** Intermittent depending on local site guidelines **Checklist Review Interval:** Regular AE and concomitant medication review at each PK draw

---

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]